THE UNITED REPUBLIC OF TANZANIA
ORGANIZATION OF HEADS OF NON-GOVERNMENT SECONDARY SCHOOLS-TANZANIA (OHONGSS-T)
MWANZA & KIGOMA REGIONS
FORM FOUR MOCK EXAMINATION MAY, 2023

041 **BASIC MATHEMATICS**

MARKING SCHEME

1. (a) Given three wooden rods of lengths $168cm, 72cm$ and $90cm$, which are cut into the same pieces. To find the greatest (GCF) length.

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*2	186	72	90
2	84	36	45
2	42	18	45
*3	21	9	45
3	7	3	15
5	7	1	5
7	7	1	1
	1	1	1

(01 Mark)

$\therefore GCF = 2 \times 3 = 6$ **(01 Mark)**

$\boldsymbol{The\ greatest\ possible\ lentgh\ (GCF) of\ wooden\ rod = 6cm}$ **(01 Mark)**

(b) Given last year's enrolment in a certain school was 500
Percentage this year is 105%
Thus, total enrolment this year = 500 +105%×500 ……**(01 Mark)**

$= 500 + \frac{105}{100} \times 500$……**(01 Mark)**

$= 500 + 525 = 1025$

$\therefore \boldsymbol{Total\ enrolment\ this\ year = 1025\ students}$……**(01 Mark)**

2. (a) Given $A = P\left(1 + \frac{R}{100}\right)^n$

To make "n" the subject

Dividing by P in both sides:

$\frac{A}{P} = \left(1 + \frac{R}{100}\right)^n$ ……**(01 Mark)**

Taking logarithm in both sides.

$\log\left(\frac{A}{P}\right) = \log\left(1 + \frac{R}{100}\right)^n$

By laws of logarithm

$\log\left(\frac{A}{P}\right) = n\log\left(1 + \frac{R}{100}\right)$ ……**(01 Mark)**

$\therefore n = \frac{\log\left(\frac{A}{P}\right)}{\log\left(1+\frac{R}{100}\right)}$ or $n = \frac{\log A - \log P}{\log\left(1+\frac{R}{100}\right)}$ ……**(01 Mark)**

(b) Given $\frac{1}{2-\sqrt{3}}$ and $\frac{1}{2+\sqrt{3}}$

Show that their sum is a rational number.

$\Rightarrow \frac{1}{2-\sqrt{3}} + \frac{1}{2+\sqrt{3}} = \frac{2+\sqrt{3}+2-\sqrt{3}}{(2+\sqrt{3})(2-\sqrt{3})}$ ……**(01 Mark)**

$= \frac{4}{2^2-\left(\sqrt{3}\right)^2}$

$= \frac{4}{4-3}$

$= \frac{4}{1} = 4$ ……**(01 Mark)**

$\therefore$ *The sum of* $\frac{1}{2-\sqrt{3}}$ and $\frac{1}{2+\sqrt{3}}$ *is* 4, *which is rational number* ……**(01 Mark)**

3. (a) Given total house holders is 160.

Let $n(C) = 95$ (*no of house holders kept cats*)

$n(D) = 80$ (*no of house holders kept dogs*)

$n(C \cap D) = ?$ (*no of house holders kept both*)

By using formula:

$Total\ n(\mu) = n(C \cup D)$

$\Rightarrow n(\mu) = n(C) + n(D) - n(C \cap D)$ ……**(01 Mark)**

$\Rightarrow 160 = 95 + 80 - n(C \cap D)$

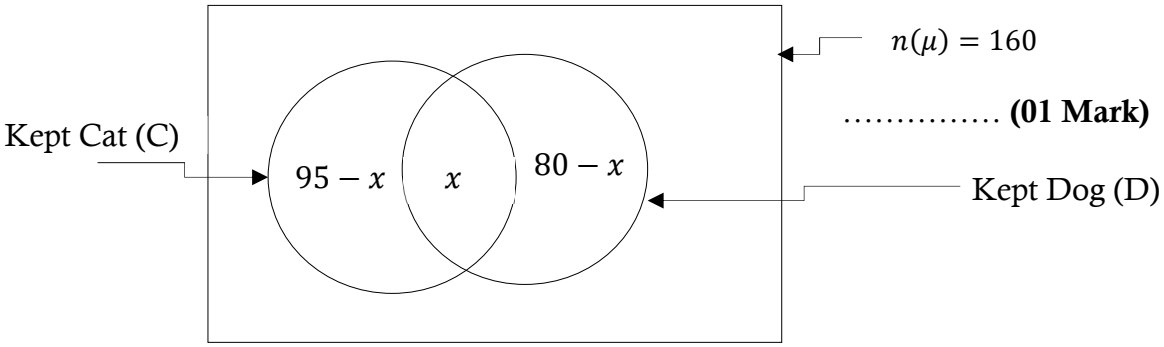
$\Rightarrow 160 = 175 - n(C \cap D)$ ……**(01 Mark)**

$\therefore n(C \cap D) = 175 - 160 = 15$

$\therefore$ 15 *house holders kept both a cat and a dog* ……**(01 Mark)**

ALTERNATIVELY

Using venn diagram:



…… **(01 Mark)**

But total $n(\mu) = n(C \cup D)$

$\Rightarrow 160 = 95 - x + x + 80 - x$ …… **(01 Mark)**

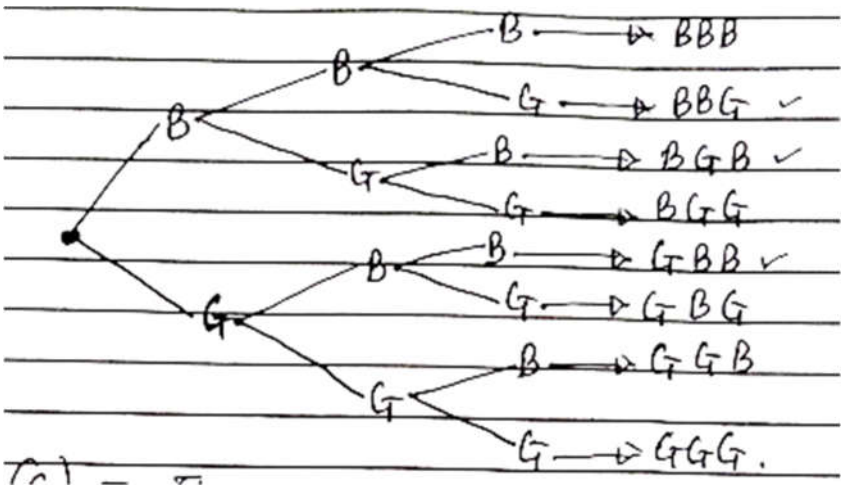
$\Rightarrow 160 = 175 - x$

$\therefore x = 175 - 160 = 15$

$\therefore 15$ *house holders kept both a cat and a dog* …… **(01 Mark)**

(b) A family has three children

By a tree diagram Results



……**(01 Mark)**

$\therefore n(S) = 8$

(i) $P(all\ girls) = \frac{1}{8}$ ……**(01 Mark)**

(ii) $P(one\ girl\ and\ two\ boys) = \frac{3}{8}$ ……**(01 Mark)**

4. (a) Given coordinates of a garden plot A (0, 0) B (3, -2) C (2, 3) and D (12, 5)

(i) To find slope of side BD.

$But\ slope = \frac{\Delta y}{\Delta x}$

$= \frac{y_2 - y_1}{x_2 - x_1}$ ……(**00$\frac{1}{2}$ Marks**)

$= \frac{5-(-2)}{12-3}$ ……(**00$\frac{1}{2}$ Marks**)

$= \frac{7}{9}$

$\therefore slope\ of\ side\ BD\ is\ \frac{7}{9}$ …… (**00$\frac{1}{2}$ Marks**)

(ii) To find coordinates of the mid point of diagonal BC. Since;

$mid - point\ (x, y) = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ …… (**00$\frac{1}{2}$ Marks**)

$= \left(\frac{3+2}{2}, \frac{-2+3}{2}\right)$ …… (**00$\frac{1}{2}$ Marks**)

$= \left(\frac{5}{2}, \frac{1}{2}\right)$

$\therefore Mid - point\ of\ diagonal\ BC\ is\ = \left(\frac{5}{2}, \frac{1}{2}\right)$ …… (**00$\frac{1}{2}$ Marks**)

(b) Given position vector $\underline{r} = (0,10) + (0.4, -0.3)$ and scalar $t = 20$. To calculate position vector of the ship.

$i.e\ \underline{r} = (0,10) + 20(0.4, -0.3)$ ……**(01 Mark)**

$= (0,10) + (2, -6)$……**(01 Mark)**

$= (2, 4)$

$\therefore\ \underline{r} = (2,4)$ ……**(01 Mark)**

5. (a) Given a nine – sided regular polygon inscribed in a circle of radius 10 cm

To find length of one side S, and perimeter P of the polygon.

- $length\ of\ one\ side\ S, = ?$

$since: S = 2r \sin\left(\frac{180°}{n}\right)\ where\ n\ is\ number\ of\ sides\ of\ the\ polygon$ …… (**00$\frac{1}{2}$ Marks**)

$But\ r = 10cm, n = 9(sides) given$

$\Longrightarrow S = 2 \times 10cm \times sin\left(\frac{180°}{9}\right)$ …… (**00$\frac{1}{2}$ Marks**)

$= 20cm \times \sin 20°$

$= 20cm \times 0.3420$

$= 6.84cm$

$\therefore Length\ of\ one\ side\ of\ the\ polygon\ is\ 6.84cm$ …… (**00$\frac{1}{2}$ Marks**)

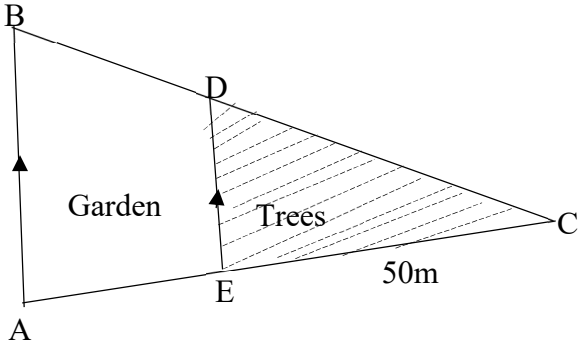
$Perimeter, P = ?$

$since\ P = ns, where\ S\ is\ length\ of\ one\ side\ of\ polygon$ …… (**00$\frac{1}{2}$ Marks**)

$\Longrightarrow P = 9 \times 6.84cm$ ……. (**00$\frac{1}{2}$ Marks**)

$\therefore Perimeter = 61.56cm.$ …… (**00$\frac{1}{2}$ Marks**)

(b) Given:



$i.e\ CE = 50m, CA = 100m, area\ of\ \Delta CDE = 72m^2$

$By\ similarity\ theorem$

$\frac{area\ ABC}{area\ CDE} = \left(\frac{CA}{CE}\right)^2$ ……. (**00$\frac{1}{2}$ Marks**)

$\Rightarrow \frac{area\ ABC}{720} = \left(\frac{100}{50}\right)^2$

$\Rightarrow area\ ABC = 720 \times \left(\frac{100}{50}\right)^2$ ……. (**00$\frac{1}{2}$ Marks**)

$= 720 \times 4 = 2880m^2$. ……. (**00$\frac{1}{2}$ Marks**)

$But\ area\ ABDE = areaABC - areaCDE$ ……. (**00$\frac{1}{2}$ Marks**)

$= 2880 - 720$

$= 2{,}160m^2$

$\therefore The\ area\ of\ portion\ ABDE\ is\ 2{,}160m^2$ ……. (**01 Marks**)

6. (a) (i) Given that a labour paid 6,000$Tshs$ for 8 hours

$\therefore His\ rate\ of\ pay = \frac{6000\ Tshs}{8\ hours}$ ……. (**00$\frac{1}{2}$ Marks**)

$= 750\ Ths\ per\ hours$ …… (**00$\frac{1}{2}$ Marks**)

(ii) Rate of grow for plant: $A = \frac{15cm}{8days} = 1.875\frac{cm}{day}$ ……. (**00$\frac{1}{2}$ Marks**)

$B = \frac{12cm}{5days} = 2.4\frac{cm}{day}$ ……. (**00$\frac{1}{2}$ Marks**)

$C = \frac{18cm}{11days} = 1.636\frac{cm}{day}$ ……. (**00$\frac{1}{2}$ Marks**)

$\therefore Plant\ C\ is\ growing\ the\ slowest.$

Because its rate of grow is 1.636 *cm per day which is the smallest of all.* (**00$\frac{1}{2}$ Marks**)

(b) Given: The number of workers to repair a road is inversely proportional to the time taken.

(i) To formulate the equation.

$i.e\ W\alpha\frac{1}{t}$ ……. (**00$\frac{1}{2}$ Marks**)

$\therefore W = \frac{k}{t}$ *where k is proportionality constant.* ……. (**01 Marks**)

(ii) From the equation above,

$w = \frac{k}{t}, Given\ w = 12(workers), t = 10\ day$

$\Rightarrow 12 = \frac{k}{10}, \quad \therefore k = 120$ ……. (**00$\frac{1}{2}$ Marks**)

$Now, t = ?\ when\ w = 30\ workers$

$\Rightarrow t = \frac{120}{w} = \frac{120}{30} = 4$ ……. (**00$\frac{1}{2}$ Marks**)

$\therefore$ 30 *workers will take* 4 *days to finish the same repair of the road* ……. (**00$\frac{1}{2}$ Marks**)

7. (a) Let x be number of sweets the teacher distributed to his/her three pupils in the baby class in the ratio 3:5:7, the smallest share being 25 sweets John got.

$\Longrightarrow smallest\ sahre = \left(\frac{3}{3+5+7}\right)x$ ……. (**01 Marks**)

$\Longrightarrow \quad 25 = \frac{3}{15}x$

$\therefore x = \left(\frac{25 \times 15}{3}\right) = 125$ ……. (**01 Marks**)

$\therefore$ *The teacher had* 125 *sweets before distributing them* ……. (**01 Marks**)

(b) (i) Trading account is an account used to determine the Gross Profit of the goods sold……. (**00$\frac{1}{2}$ Marks**)

(ii) Opening stoke ……..Tsh 8,000.

Add: Purchases…… Tsh 25,000.

Cost of goods available for sale……Tsh 33,000 ……. (**00$\frac{1}{2}$ Marks**)

Less: Closing stock ……Tsh 5,000

Cost of goods sold ……Tsh 28,000……. (**00$\frac{1}{2}$ Marks**)

$\therefore Gross\ profit = sales - cost\ of\ goods\ sold$

$= 38{,}000 - 28{,}000$

$= Tsh = 10{,}000$ ……. (**00$\frac{1}{2}$ Marks**)

But given expenses:

- Electricity bill ……Tsh 4,000
- Discount allawed……Tsh 2,000
- Total ……Tsh 6,000…. … … (**00$\frac{1}{2}$ *Marks***)

$\therefore Net\ profit = Gross\ profit - Expenses$

$= 10{,}000 - 6{,}000$

$= Tsh\ 4{,}000$ …. … … … … … … … … … … … … … … (**00$\frac{1}{2}$ *Marks***)

8. (a) Given 40 000, 45 000, 50 000, … which is A.P

Let $A_1 = 40\,000, d = 5000. A_n = ?$

But $A_n = A_1 + (n-1)d$ …. … (**01 *Marks***)

$= 40000 + (n-1) \times 5000$

$= 40000 + 5000n - 5000$ …. … … … … … … … … … … … … … … … … … (**01 *Marks***)

$= 35000 + 5000n$

$\therefore$ *In the* n^{th} *year of the, the woman saves* $35000 + 5000n$, *where n is the number of year of her job* …. … (**01 *Marks***)

(b) Given $6 + 6.6 + 7.26 + \cdots$ *which is G.P*

let $G_1 = 6, G_2 = 6.6$

$\therefore r = \frac{G_2}{G_1} = \frac{6.6}{6} = 1.1$ … (**01 *Marks***)

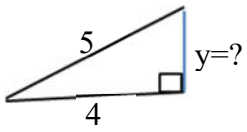
$S_{12} = ?$

$Since; S_n = \frac{G_1(r^n - 1)}{r=1}$ …. … (**01 *Marks***)

$\Longrightarrow S_{12} = \frac{6(1.1^{12} - 1)}{1.1 - 1.0} = 128.3$

$\therefore$ *Total distance run after* 12 *days is* 128.3 *km* … … … … … … … … … (**01 *Marks***)

9. (a) Given: $\cos A = \frac{4}{5}$ *i.e*



By pythagoras theorem

$y = \sqrt{5^2 + 4^2} = \sqrt{25 - 16} = \sqrt{9} = 3$ … … … … … … … … … … … … … … (**00$\frac{1}{2}$ *Marks***)

(i) To find $\sin A$

$\therefore \sin A = \frac{Opposite}{Hypotenus} = \frac{3}{5}$

$\therefore \sin A = \frac{3}{5}$ …. … (**01 *Marks***)

(ii) $To\ find\ value\ of\ \frac{\tan A-\cos A}{1+\cos A}$

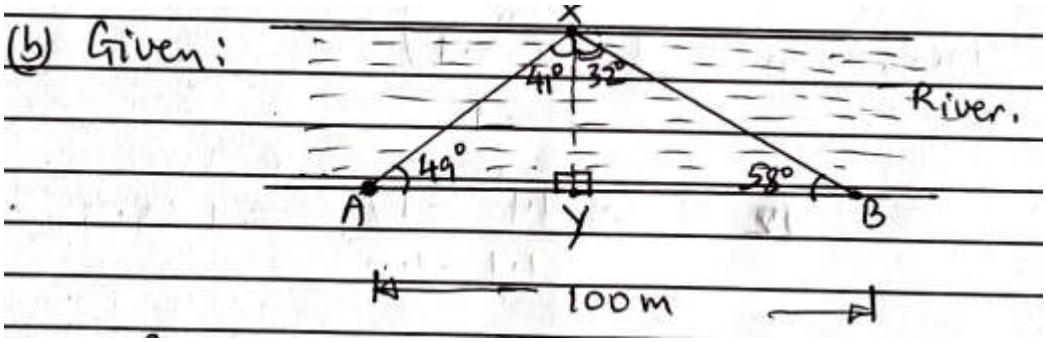
$But:\ \tan A = \frac{Opposite}{Adjacent} = \frac{3}{4}$

$\therefore \frac{\tan A-\cos A}{1+\cos A} = \frac{\frac{3}{4}-\frac{4}{5}}{1+\frac{4}{5}}$ …… (**00$\frac{1}{2}$ Marks**)

$= \frac{\frac{15-16}{20}}{\frac{5+4}{5}} = \frac{-1}{20} \div \frac{9}{5}$

$= \frac{-1}{20} \times \frac{5}{9} = \frac{-1}{36}$

$\therefore \frac{\tan A - \cos A}{1 + \cos A} = \frac{-1}{36}$ …… (**01 Marks**)



To find width $\overline{XY}$ of the river

By sine rule in ΔABX

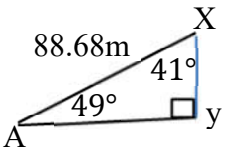
$\frac{\sin(41° + 32°)}{AB} = \frac{\sin 49°}{BX} = \frac{\sin 58°}{AX}$ …… (**00$\frac{1}{2}$ Marks**)

$for\ \frac{\sin 73°}{AB} = \frac{\sin 58°}{AX}$

$\Longrightarrow \frac{\sin 73°}{100} = \frac{\sin 58°}{AX}$

$\therefore AX = 88.68m$ …… (**00$\frac{1}{2}$ Marks**)

$also\ for \Delta AXY$



…… (**01 Marks**)

$\Rightarrow XY = 88.68 \sin 49°$

$= 66.93m$

$\therefore Width\ XY\ of\ the\ river\ is\ 66.93m\ width.$ …… (**01 Marks**)

10. (a) *Letx be age of the man, y be age of his son.*

Given that $x = 4y$ ……. (**00$\frac{1}{2}$ Marks**)

Also $xy = 196$ (*product of their ages*) …… (**00$\frac{1}{2}$ Marks**)

Thus, by substitution method,

$(4y)y = 196$

$4y^2 = 196$

$y^2 = 49$ ……. (**01 Marks**)

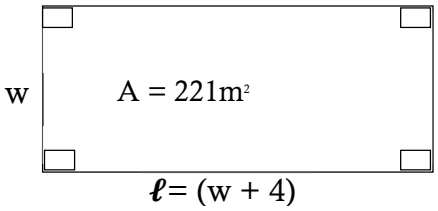
$y = 7$ *years and* $x = 4y = 4(7) = 28$ *years.*

∴ *The son is* 7 *years old.* …… (**01 Marks**)

(b) Let ℓ be length of a classroom floor given.

W be width.

Given that ℓ = w +4 …… (**00$\frac{1}{2}$ Marks**)



But $A = \ell w$

$\Rightarrow 221 = (w + 4)w$ ……. (**00$\frac{1}{2}$ Marks**)

$\Rightarrow 221 = w^2 + 4w$

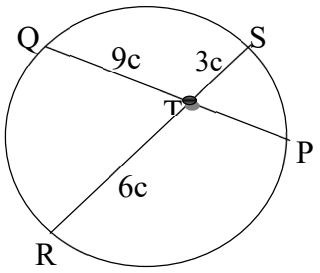
∴ $w^2 + 4w - 221 = 0$ *which is quadratic equation.* ……. (**01 Marks**)

Solving the quadratic equation by any method

$w = 13$ *or* -17 ……. (**00$\frac{1}{2}$ Marks**)

∴ *The length of the floor is* 17m *and its width is* 13m. ……. (**00$\frac{1}{2}$ Marks**

11. (a) Given



But $PT \times QT = RT \times ST$ (*for intersecting chords*)

$\Rightarrow PT \times 9 = 6 \times 3$ ……. (**01 Marks**)

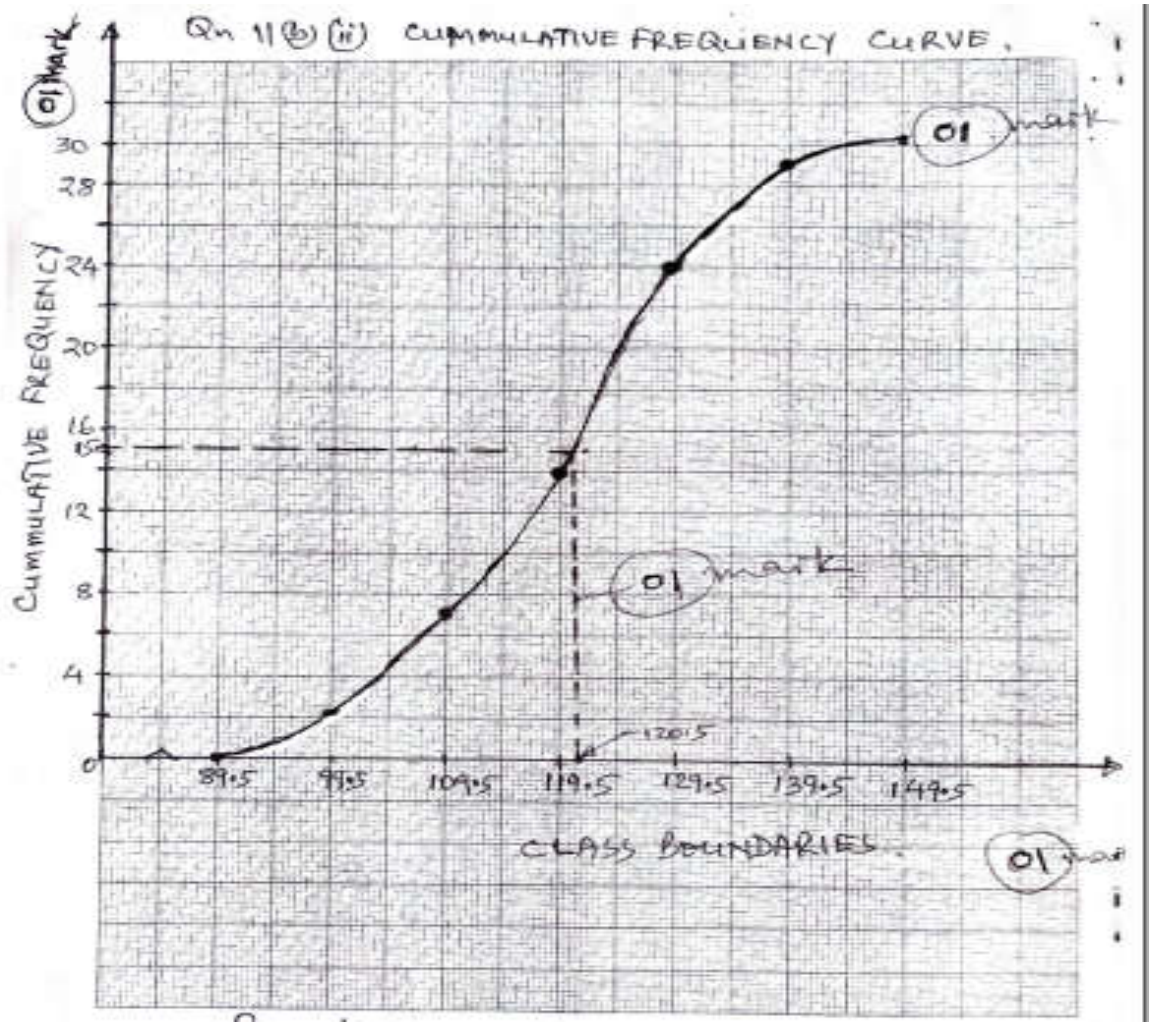
∴ $PT = \frac{6\times3}{9} = 2cm$ ……. (**01 Marks**)

∴ $TP = 2cm$ ……. (**01 Marks**)

(b) (i) Frequency distribution table for heights of 30 children, including class boundaries or limits and a cumulative frequency.

Heights (cm)	Class boundaries (Real limit)	Class mark (x)	F	c.f
90 – 99	89.5 – 99.5	94.5	2	2
100 – 109	99.5 – 109.5	104.5	5	7
110 – 119	109.5 – 119.5	114.5	7	14
120 – 129	119.5 – 129.5	124.5	10	24
130 – 139	129.5 – 139.5	134.5	5	29
140 – 149	139.5 – 149.5	144.5	1	30

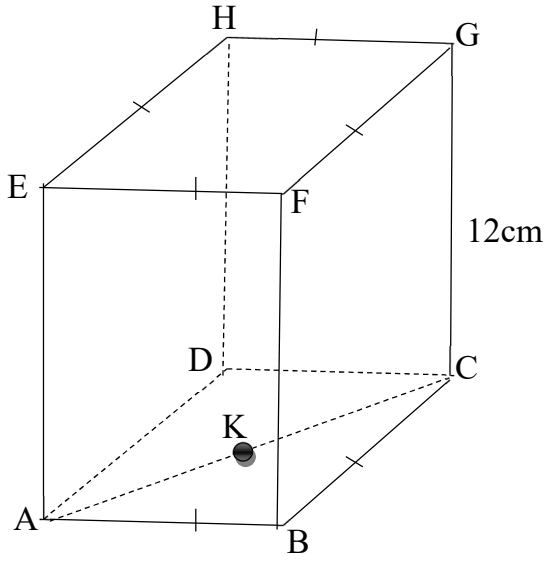
(**01 *Marks***) (**01 *Marks***)



(**04 *Marks***)

From the graph (curve)the estimate of median height is 120.5*cm* (**01 *Marks***)

12. (a) Given a cuboid having a square base of side $AB = BC, diagonal\ AC = 5cm$



(i) To find length AG

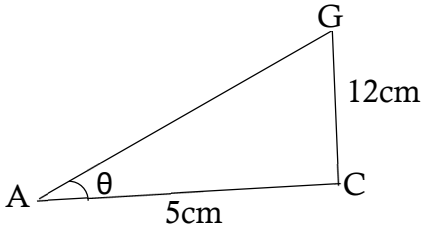
By Pythagoras theorem

$AG = \sqrt{AC^2 + CG^2}$ …. … (**01 *Marks***)

$= \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13.$

$\therefore length\ AG = 13cm.$ …. … (**01 *Marks***)

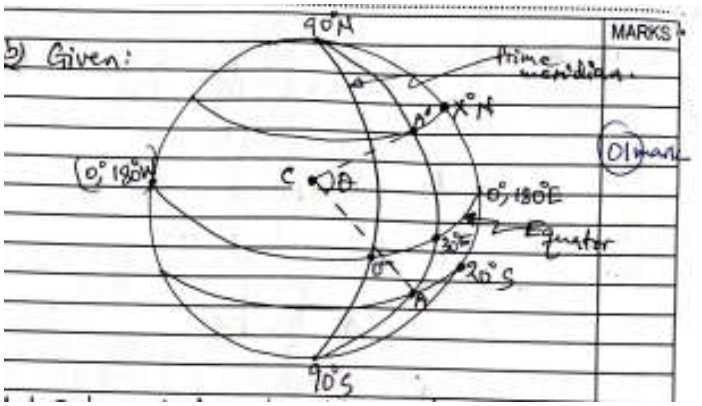
(ii) To calculate angle between AG and base ABCD. In ΔACG



…. … … … …. … … (**01 *Marks***)

$\Longrightarrow \tan(C\hat{A}G) = \frac{CG}{AC}$ … … … … … … … … … … … … … … … … … … …

(b)



(**01 *Marks***)

Let θ be central angle subtended by arcAA' along the meridian.

But $\theta = 20° + x°$ …. … (**01 *Marks***)

since distance $AA' = \frac{2\pi R_e \theta}{360°}$ … (**01 *Marks***)

Given distance $AA' = 4000km$ *and* $R_e = 6370km$

$\Rightarrow 4000 = \frac{2\times 3.14\times 6370\theta°}{360°}$

$\therefore \theta = \frac{4000\times 360}{2\times 3.14\times 6370} = 36°$ … (**01 *Marks***)

But $\theta = 20° + x°$

$\Rightarrow 36° = 20° + x°$

$\therefore x = 36° - 20° = 16°$

$\therefore$ *The new latitude is* $16°N$ *along the same longitude of* $30°E$ … … (**01 *Marks***)

or simply the new position of the plane is $(16°N, 30°E)$

13. (a) Let x be cost of a shirt, y be cost of pair of shorts…. … … … … … … … … (**00$\frac{1}{2}$ *Marks***)

Given: $\begin{cases} 5x + 2y = 27{,}000 \ldots . . eqn(i) \\ 2x + 6y = 29{,}000 \ldots . eqn(ii) \end{cases}$. … (**01 *Marks***)

(i) To find cost of a shirt and that of a pair of shorts

$3\begin{cases} 5x + 2y = 27{,}000 \\ 1\begin{cases} 2x + 6y = 29{,}000 \end{cases} \end{cases}$. … … … … … … … … … … … … … … … … … … …

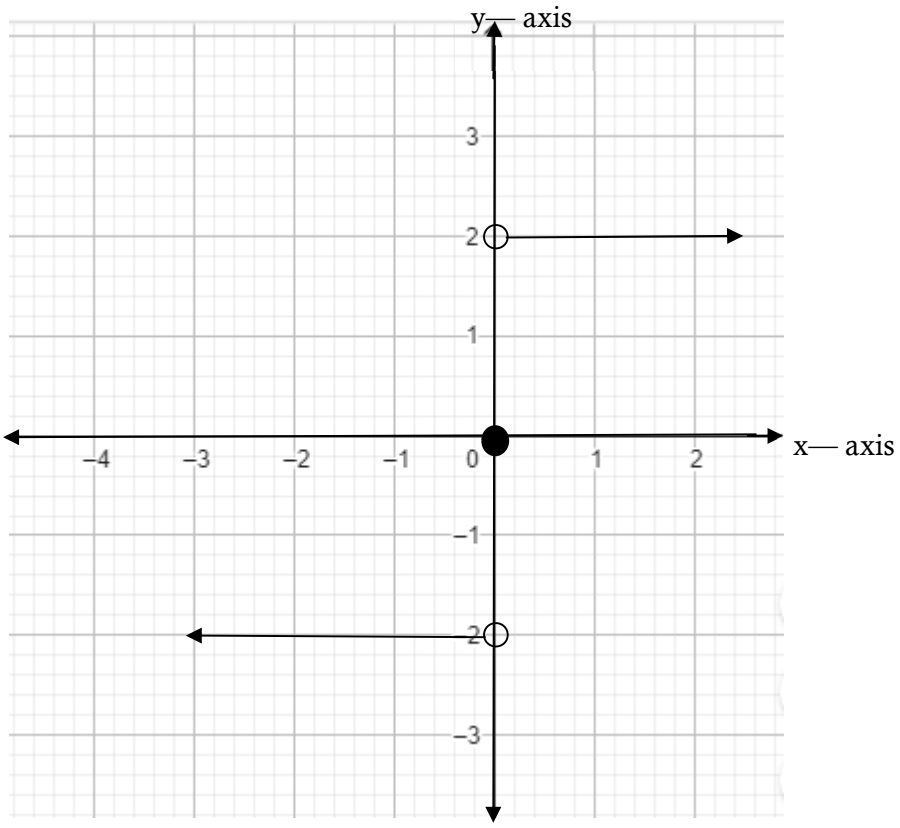
(ii) Total cost of 8 shirts and 5 pairs of shorts

$= 8x + 5y$ ……. (**$00\frac{1}{2}$ *Marks***)

$= 8(4000) + 5(3500) = 49{,}500\ Tshs$ …… (**01 *Marks***)

(b) To draw the graph of $f(x) = \begin{cases} -2\ if x < 0 \\ 0\ if\ x = 0 \\ 2\ if\ x\ > 0 \end{cases}$

Graph



- *Each circled point* $= 00\frac{1}{2} \times 3 = 01\frac{1}{2} Marks$
- *Each axis* $00\frac{1}{2} \times 2 = 01 Marks$
- ***Total marks*** $= \mathbf{02\frac{1}{2}}$

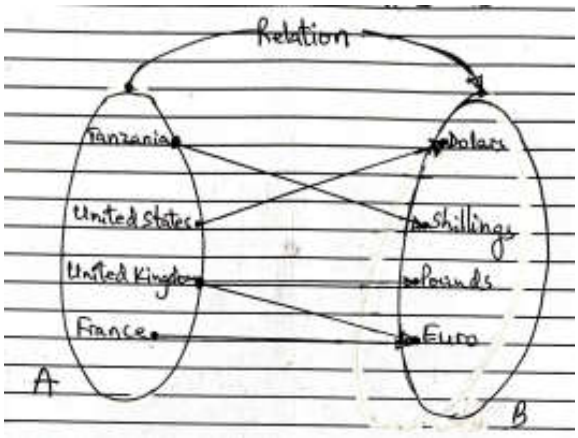
Hence From the graph

(i) $f(7) = 2$ …… (**$00\frac{1}{2}$ *Marks***)

(ii) *Range of* $f(x) = \{-2, 0, 2\}$

(*Each value of range* (**$00\frac{1}{2} \times 3 =$ $01\frac{1}{2}$ *Marks***)

14. (a) (i) The arrow diagram representing the relation between A and B is



…… (**00$\frac{1}{2}$ *Marks*)

(ii) This relation is not a function …… (**00$\frac{1}{2}$ *Marks*)
Because one member of set A does not correspond to one and only one member of set B. (**00$\frac{1}{2}$ *Marks*)
OR

An element in set A (United Kingdom) corresponds to two different elements in set B (Pounds and Euro)

(b) Let x be units of food type 1, y be units of food type 2.

Objective function: $f(x, y) = x + y$ to minimize …… (**00$\frac{1}{2}$ *Marks*)

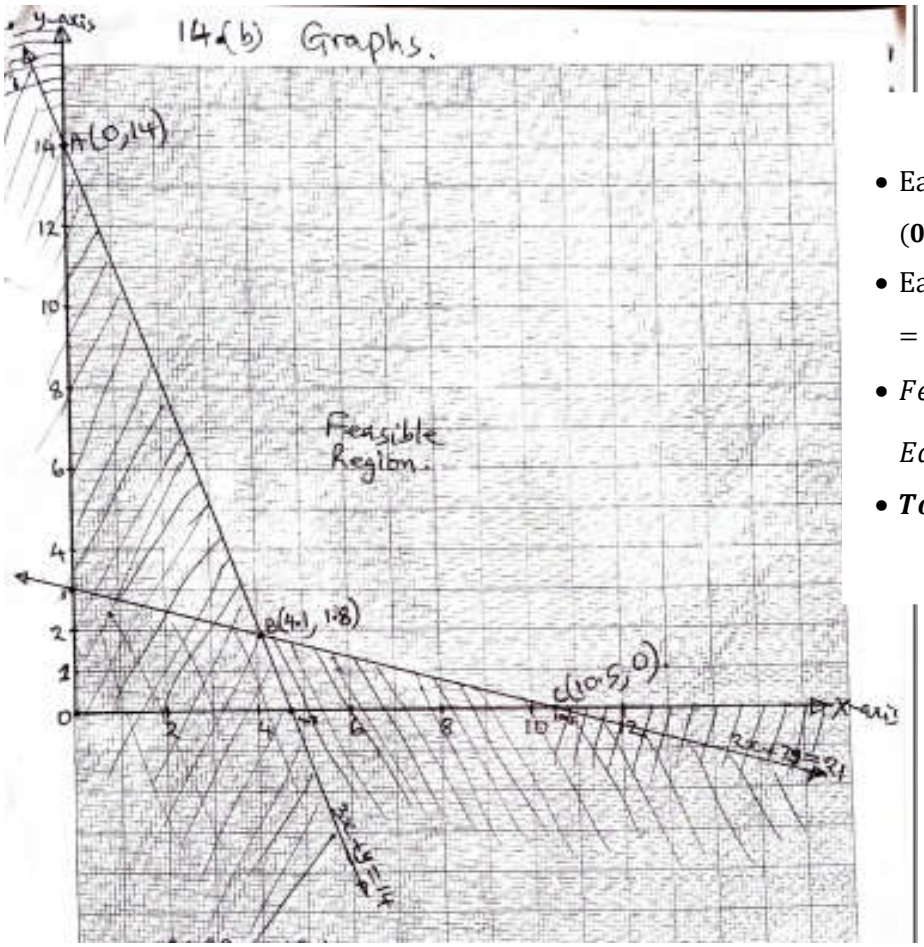
Constraints/ inequalities

$vitamine\ A$: $3x + y \geq 14$ …… (**00$\frac{1}{2}$ *Marks*)

$Vitamine\ B$: $2x + 7y \geq 21$ …… (**00$\frac{1}{2}$ *Marks*)

$x \geq 0$
$y \geq 0$ } …… (**00$\frac{1}{2}$ *Marks*)

Graph



Marking

- Each correct straight $(\mathbf{00\frac{1}{2} \times 2 = 01\ Marks})$
- Each correct corner point $= \mathbf{00\frac{1}{2} \times 3 = 01\frac{1}{2}\ Marks}$
- *Feasible region* $= \mathbf{00\frac{1}{2}\ Marks}$
  Each axis $(\mathbf{00\frac{1}{2} \times 2 = 01\ Marks})$
- $\boldsymbol{Total\ marks\ for\ graph\ 04 Marks}$

To get corner points of the feasible region in the objective function.

Corner points	Objectives function f (x, y) = x + y
A (0, 14)	f (0, 14) = 0 + 14 = 14 $(\mathbf{00\frac{1}{2}\ Marks})$
B (4.1, 1.8)	f (4.1, 1.8) = 4.1 + 1.8 = 5.9 $(\mathbf{00\frac{1}{2}\ Marks})$
C (10.5, 0)	f (10.5, 0.0) = 10.5 + 0 = 10.5 $(\mathbf{00\frac{1}{2}\ Marks})$

Therefore, food type 1 should be 4:1 units and food type 2 should be 1:8 units to fit the doctor's prescription.

$\left(\boldsymbol{Each\ value\ 00\frac{1}{2} \times 2 = 01\ Marks}\right)$